

Indian Institute of Technology Kharagpur

AUTUMN Semester, 2026–2027

COMPUTER SCIENCE AND ENGINEERING

Computer Organization and Architecture Laboratory

Verilog Assignment 6

August 2026

***Instructions:** This is a group assignment. Each group must implement the complete Verilog design, write a suitable testbench, program the FPGA board, and demonstrate the working design on the board to a Teaching Assistant for evaluation. The behavioral simulation must also be demonstrated to the Teaching Assistant.*

Question 1: 4-bit Sequential Booth Multiplier with Visible Iterations

1. Objective

Design and implement a **4-bit signed sequential multiplier** using Booth's Multiplication Algorithm.

Both operands are 4-bit signed two's-complement numbers, and the final product is an 8-bit signed two's-complement number.

The multiplication must be carried out **iteratively**. One Booth iteration must be performed on every rising edge of a **slow clock** generated from the FPGA board clock. The slow-clock technique demonstrated in the laboratory may be used so that the progress of the multiplication can be observed on the board LEDs.

A purely combinational implementation using the Verilog multiplication operator is not permitted.

2. Top-Level Interface

Create a top-level module named

`booth_multiplier_4bit`

with the following interface:

Signal	Width	Description
<code>clk</code>	1 bit	FPGA board clock
<code>reset</code>	1 bit	Active-high reset
<code>start</code>	1 bit	Starts a new multiplication
<code>multiplicand</code>	4 bits	Signed two's-complement multiplicand
<code>multiplier</code>	4 bits	Signed two's-complement multiplier
<code>display_sel</code>	1 bit	Selects internal-state or final-result display
<code>led</code>	16 bits	Output connected to the board LEDs

A suggested board mapping is:

- `multiplicand[3:0]`: four slide switches;
- `multiplier[3:0]`: four slide switches;
- `display_sel`: one slide switch;
- `reset`: one pushbutton;
- `start`: one pushbutton; and
- `led[15:0]`: the user LEDs.

3. Booth Registers

The sequential Booth multiplier must maintain the following internal registers:

Register	Width	Purpose
A	5 bits	Accumulator used during addition/subtraction
M	5 bits	Sign-extended multiplicand
Q	4 bits	Multiplier register
Q_{-1}	1 bit	Previous least-significant bit of Q
<code>count</code>	3 bits	Number of Booth iterations completed

The 4-bit multiplicand must be sign-extended before being stored in M .

The accumulator A and multiplicand register M are specified as 5-bit signed quantities so that the complete 4-bit signed input range, including -8 , can be handled safely during intermediate addition/subtraction.

4. Initialization

When `reset` is asserted, the circuit must return to its initial state.

When `start` is asserted after reset, the current operand values must be captured and the Booth registers initialized as

$$A \leftarrow 0,$$

$$M \leftarrow \text{sign-extended multiplicand},$$

$$Q \leftarrow \text{multiplier},$$

$$Q_{-1} \leftarrow 0,$$

and

$$\text{count} \leftarrow 0.$$

The multiplication must then proceed for exactly four Booth iterations.

5. Booth Decision

At the beginning of every iteration, examine the two-bit value

$$\{Q_0, Q_{-1}\},$$

where Q_0 is the least-significant bit of Q .

The required operation is:

$\{Q_0, Q_{-1}\}$	Operation
00	No addition or subtraction
01	$A \leftarrow A + M$
10	$A \leftarrow A - M$
11	No addition or subtraction

After this operation, an arithmetic right shift must be applied to the combined Booth register

$$\{A, Q, Q_{-1}\}.$$

The sign bit of A must be preserved during the arithmetic right shift.

Finally,

$$\text{count} \leftarrow \text{count} + 1.$$

6. Booth's Algorithm Flowchart

The following flowchart summarizes the Booth iteration used in this assignment.

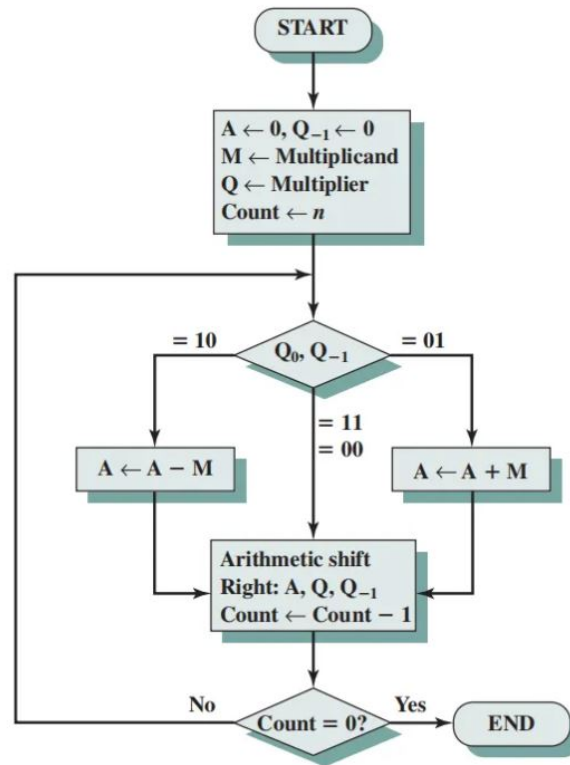


Figure 1: Booth's Multiplication Algorithm

For this assignment,

$$n = 4,$$

so the loop is executed exactly four times.

7. Slow Clock and Visible Execution

The FPGA board clock is too fast for the individual Booth iterations to be observed directly.

Use the slow-clock generation technique demonstrated in the laboratory. A counter may be incremented using the board clock, and a suitable higher-order counter bit may be used as

`slow_clk`.

For example,

Board Clock $\longrightarrow$ Clock-Divider Counter $\longrightarrow$ `slow_clk`

may be used.

One complete Booth iteration must occur on each rising edge of `slow_clk`.

The selected clock division must be slow enough for the four successive iterations to be visibly distinguished during the FPGA-board demonstration.

Students may temporarily connect the slow-clock signal to an LED to verify visible LED blinking before integrating the Booth multiplier.

8. Formation of the Final Product

After four iterations, the final signed 8-bit product must be taken from

$$\text{product}[7:0] = \{A[3:0], Q[3:0]\}$$

where $A[4]$ is the sign-extension bit of the internal accumulator.

Once the fourth iteration is complete:

- a **done** condition must be asserted internally;
- the product must remain stable; and
- no further Booth iterations must be performed until the circuit is reset and started again.

9. LED Display Requirement

The LEDs must allow the TA to observe both the progress of the Booth algorithm and the final product.

Use `display_sel` as follows.

Internal-State Display: `display_sel = 0`

A suggested mapping is:

LEDs	Displayed Signal
LED[4:0]	$A[4:0]$
LED[8:5]	$Q[3:0]$
LED[9]	Q_{-1}
LED[12:10]	count
LED[13]	Done indication
LED[15:14]	0

With this display, the internal state should visibly change once per slow-clock edge.

Final-Result Display: `display_sel = 1`

Use:

LEDs	Displayed Signal
LED[7:0]	Final signed 8-bit product
LED[8]	Done indication
LED[15:9]	0

Equivalent LED mappings may be used if clearly documented in the submission and demonstrated to the TA.

10. Worked Example

Consider

$$\text{multiplicand} = 0011 = +3$$

and

$$\text{multiplier} = 1110 = -2.$$

The expected result is

$$3 \times (-2) = -6.$$

The 8-bit two's-complement representation of -6 is

$$\boxed{11111010}.$$

At initialization,

$$A = 00000,$$

$$M = 00011,$$

$$Q = 1110,$$

and

$$Q_{-1} = 0.$$

The circuit must then execute four Booth iterations. During each iteration, students should be able to observe:

1. the current pair $\{Q_0, Q_{-1}\}$;
2. whether $A + M$, $A - M$, or no arithmetic operation is selected;
3. the arithmetic right shift of $\{A, Q, Q_{-1}\}$; and
4. the increment of the iteration count.

After the fourth iteration,

$$\text{product} = 11111010$$

and the done indication must be asserted.

11. Required Test Cases

At minimum, verify the following signed input combinations:

Multiplicand	Multiplier	Expected Product
3	2	6
3	-2	-6
-3	2	-6
-3	-2	6
7	3	21
-8	1	-8
7	-8	-56
-8	-8	64

Additional test cases may be used during evaluation.

12. Testbench Requirement

Write a behavioral testbench for the multiplier.

The testbench must verify:

- reset and initialization;
- correct loading of signed operands;
- correct Booth decision for 01, 10, 00, and 11;
- exactly four iterations;
- arithmetic right shifting;
- correct positive and negative products; and
- stable output after completion.

The waveform should show at least

`clk, reset, start, multiplicand, multiplier,`

and the internal signals

`A, Q, Q-1, count`

together with the final product/done indication.

A much smaller clock-divider value may be used in the testbench to avoid an unnecessarily long simulation.

13. Design Restrictions

The following requirements must be satisfied:

1. The Verilog multiplication operator `*` must not be used to compute the product.
2. The design must be sequential and iterative. The complete multiplication must not be performed in a single clock cycle.
3. Exactly one Booth iteration must be performed per slow-clock edge.

4. Sequential registers must be updated using non-blocking assignments.
5. The shift of $\{A, Q, Q_{-1}\}$ must be an **arithmetic right shift**, preserving the sign of A .
6. The final product must remain unchanged after completion until the design is reset for another multiplication.

14. Mandatory FPGA-Board Demonstration

Demonstration on the FPGA board is mandatory.

Each group must demonstrate the complete Booth multiplier to a Teaching Assistant.

The demonstration must include:

1. Setting signed multiplicand and multiplier values using the board switches.
2. Resetting and starting the multiplier.
3. Showing the internal Booth state changing on the LEDs for each slow-clock iteration.
4. Showing that the iteration count progresses through the four Booth steps.
5. Switching to the final-result display and showing the correct signed 8-bit product.
6. Demonstrating at least one positive-result case and one negative-result case.

The TA may request different operand values during the live demonstration.

15. Submission

Submit the Verilog source file(s), testbench, and XDC constraint file in a single zipped folder named

`VerilogAssignment5_{Group_Number}.zip`.

Each source file must contain the group number, names and roll numbers of all group members, assignment number, and date of submission.